F1 StrategyIQ

Complete Input → Processing → Output → Visuals Specification

Purpose: Define exactly what users can enter/select, what the system processes, what it produces, and which visualizations appear in the interactive dashboard.

1. The Project Is Interactive

F1 StrategyIQ will not be a text-only application. The main deliverable is an interactive Streamlit dashboard backed by FastAPI, PostgreSQL, ML models, a race simulator and an optimization/decision engine.

User interaction: select a race/driver → choose or modify a scenario → run analysis → see predictions, strategy comparisons, probability distributions and visual recommendations → change inputs → results update.

2. Input Types

Input Category	Examples	Where Used
Race Selection	Season, GP/circuit, race/session	Database retrieval, race analysis
Driver Selection	Driver, constructor/team	Performance and ML features
Tyre Inputs	Starting compound, tyre age, stint	Tyre model, strategy simulation
Race-State Inputs	Current lap, position, gap ahead/behind, recent pace	Dynamic Pit-Wall
Pit Inputs	Pit now, pit lap, number of stops, pit duration/loss	Strategy simulation/optimization
Weather Inputs	Track temp, air temp, rainfall, wet/dry scenario	Pace, tyre and scenario models
Race-Event Inputs	Safety Car, VSC, competitor pit, track status	Dynamic decisions
Strategy Inputs	Allowed compounds, 1/2/3-stop strategy, pit window	Strategy optimizer
Risk Inputs	Conservative / Balanced / Aggressive	Multi-objective strategy selection
What-If Inputs	Stay out 3/5 laps, pit earlier/later, change tyre	Scenario comparison

3. Input Type A — Race & Driver Selection

User selects: Season → Grand Prix → Driver → optional comparison driver/team.

Example: 2024 → Bahrain GP → Driver A.

Outputs: race overview, driver statistics, position history, lap-time trend, tyre stints and pit history.

Visuals: race position line chart, lap-time line chart, stint timeline, driver KPI cards.

4. Input Type B — Tyre & Stint

Inputs: compound, tyre age, stint number, starting tyre and candidate replacement tyre.

ML output: predicted lap time and expected degradation/pace loss.

Decision output: estimated remaining useful tyre window and whether the current tyre is becoming strategically unattractive.

Visuals: tyre degradation curve, compound comparison chart, tyre-age vs lap-time scatter/line plot, projected tyre-life indicator.

5. Input Type C — Current Race State

Inputs: current lap, position, tyre, tyre age, gap ahead, gap behind, recent pace, remaining laps, weather, Safety Car/VSC and estimated degradation.

Processing: current state is passed to prediction models and the strategy engine.

Outputs: PIT NOW / STAY OUT / optional PIT WINDOW recommendation, confidence, projected position and explanation.

Visuals: live-style race-state KPI cards, gap-to-car chart, pit-window timeline, recommendation panel, confidence gauge/indicator.

6. Input Type D — Strategy Selection

Inputs: candidate tyre sequence, number of stops, possible pit laps and starting tyre.

Processing: every feasible strategy is simulated over the remaining race.

Outputs: expected race time, expected finishing position, podium probability, win probability, risk and strategy ranking.

Visuals: ranked strategy table, expected race-time bar chart, finishing-position distribution, strategy timeline.

7. Input Type E — What-If Scenario

This is one of the most interactive features. The user changes one assumption and the system re-runs the relevant simulation.

User changes	System recalculates	Visual output
Pit now	Fresh-tyre pace + pit loss + remaining laps	Expected finish/time comparison
Stay out 3 laps	Older-tyre degradation + track position	Strategy outcome comparison
Stay out 5 laps	Longer stint + changed pit window	Position/probability comparison
Change tyre	Compound performance + degradation	Tyre performance chart
Safety Car ON/OFF	Pit-loss scenario + track-position effects	Before/after strategy chart
Rain/wet scenario	Weather/tyre assumptions	Compound suitability + outcome distribution
Risk preference	Change objective weighting	Strategy ranking changes

Example: User clicks “STAY OUT +5 LAPS.” The dashboard compares PIT NOW vs STAY +5 and shows expected race time, expected position, podium probability and risk.

8. Input Type F — Dynamic Pit-Wall / Race Replay

Because real professional live timing is extremely fast and proprietary, the academic project will demonstrate dynamic decision-making through a **historical race replay**. The race is advanced lap-by-lap, with major events triggering additional re-evaluation.

At each update: current race state → prediction → strategy alternatives → decision → next historical race state → re-evaluate.

Race State	Example
Lap	28 / 57
Position	P5
Tyre	Medium
Tyre age	18 laps
Gap ahead / behind	2.1s / 1.4s
Recent pace	1:34.8
Degradation	High
Safety Car	ON

Decision output: PIT NOW — Confidence 84% — Reason: high degradation + Safety Car + favorable pit-loss conditions.

Visuals: race replay controls, current-lap indicator, position chart, gap chart, tyre timeline, pit-window timeline, recommendation card and decision history.

9. ML Model Outputs

Model	Main Inputs	Outputs	Visuals
Lap-Time Regression	Driver, circuit, compound, tyre age, lap, recent pace, gaps, fuel	Predicted lap time; error/uncertainty	Actual vs predicted lap-time chart; residual/error plot
Tyre Degradation	Compound, tyre age, circuit, pace, weather, driver	Degradation rate; projected pace loss; tyre window	Degradation curve; compound comparison
Pit Decision Classifier	Lap, position, tyre age, gaps, pace, degradation, Safety Car	PIT NOW / STAY OUT / optional window; confidence	Decision timeline; confidence indicator; feature explana

10. Strategy Simulator Outputs

For each candidate strategy, the simulator produces a projected outcome rather than a single text answer.

Metric	Meaning
Expected race time	Average simulated race duration under the model assumptions.
Expected finish	Average finishing position across simulations.
Podium probability	Share of simulations ending P1–P3.

Metric	Meaning
Win probability	Share of simulations ending P1.
Risk / variability	Spread of simulated outcomes or risk score.
Pit-stop schedule	Recommended/assumed pit laps and tyre sequence.

Visuals: strategy ranking table, bar chart of expected race time, podium/win probability chart, finishing-position histogram, Monte Carlo distribution and tyre-stint timeline.

11. Complete Dashboard Visual Set

Visualization	What it communicates
KPI Cards	Current lap, position, tyre, tyre age, gaps, predicted pace and recommendation.
Race Position Chart	How position changes over the race.
Lap-Time Chart	Actual pace and ML-predicted pace over laps.
Tyre Degradation Curve	How pace changes with tyre age and where degradation accelerates.
Compound Comparison	Relative expected performance of Soft/Medium/Hard.
Pit-Window Timeline	Recommended and alternative pit windows.
Strategy Comparison Bar Chart	Expected race time or score across candidate strategies.
Monte Carlo Distribution	How simulated finishing positions are distributed.
Podium/Win Probability	Probability comparison between strategies.
Gap Chart	Gap ahead/behind and strategic traffic context.
Stint Timeline	Tyre sequence and pit stops across the race.
What-If Comparison	Before/after outcome when a user changes a decision.
Decision History	How PIT NOW/STAY OUT recommendations changed through the replay.

12. Example Full User Journey

Step 1 — Select race: 2024 Bahrain GP → Driver A.

Step 2 — Dashboard loads: race position, lap times, tyre stints and historical performance visuals.

Step 3 — Select current state: Lap 28, P5, Medium, tyre age 18, gap ahead 2.1s, gap behind 1.4s.

Step 4 — Prediction: model predicts increasing tyre degradation and future lap-time loss.

Step 5 — Strategy simulation: compare M→H, M→M→H and other feasible strategies.

Step 6 — Optimization: M→M→H ranks highest under the selected risk preference.

Step 7 — Pit-Wall: Safety Car occurs → system re-evaluates → PIT NOW becomes preferred.

Step 8 — What-if: user tests staying out for five laps and sees the projected decline in finish/podium probability.

Step 9 — Replay: advance to the next lap and observe whether the recommendation changes again.

13. Backend Flow for Every Interactive Action

Streamlit UI → FastAPI endpoint → input validation → feature generation → PostgreSQL retrieval → ML model(s) → simulator/optimizer → structured result → Streamlit visual update.

The API therefore returns structured values, not only text. Example response fields can include predicted lap time, degradation, recommended action, confidence, expected position, strategy score and probability metrics.

14. Example Structured Output

Field	Example
recommended_action	PIT_NOW
confidence	0.84
predicted_lap_time_sec	94.82
degradation_sec_per_lap	0.09
recommended_compound	MEDIUM
recommended_pit_lap	28
expected_finish	P3
podium_probability	0.72
win_probability	0.31
risk_level	MEDIUM

The frontend converts these structured outputs into charts, KPI cards, timelines and recommendation panels. Therefore the user gets both **numbers and visual explanations**, not just a paragraph.

15. What Each Project Layer Produces

Layer	Main Output
Data Layer	Clean historical F1 database and analysis-ready feature tables.
EDA/Statistics	Trends, correlations, tyre/race insights and visual analytics.
ML Layer	Predicted lap time, tyre degradation and pit-decision probabilities.
Simulation	Possible race outcomes under alternative strategies.
Optimization	Ranked strategies and preferred strategy.
Dynamic Pit-Wall	Current action recommendation + confidence + reasons.
API	Structured JSON results connecting backend intelligence to frontend.
Frontend	Interactive charts, controls, tables, timelines and what-if analysis.

16. Final Project Output

The final product should feel like a **mini F1 strategy analytics platform**: a user selects a race and scenario, explores historical data, receives ML predictions, compares simulated strategies, runs what-if scenarios and replays the race to observe dynamic pit-wall decisions.

Core principle: Analyze → Predict → Simulate → Optimize → Decide → Re-evaluate.

Important: all probabilities and recommendations should be presented as model-based estimates under assumptions, not guarantees about real-world race outcomes.